

POC-3 : Secret: Volume Mount + Env Var Reference

Steps:

1 . Create Secret db-credentials with keys username and password.

```
POC-3-SECRET-1 > ! secret.yaml > {} stringData > abc password
```

```
1  apiVersion: v1
2  kind: Secret
3  metadata:
4    name: db-credentials
5  type: Opaque
6  stringData:
7    username: admin
8    password: MyDBPassword123
```

2 . Create pod secret-inspector (toolbox image), that:

- mounts db-credentials as a volume at /etc/app/secrets
- injects username as env var DB_USERNAME via secretKeyRef

```
POC-3-SECRET-1 > ! pod.yaml > {} spec > [] containers > {} 0 > abc imagePullPolicy
```

```
1  apiVersion: v1
2  kind: Pod
3  metadata:
4    name: secret-inspector
5  spec:
6    containers:
7      - name: toolbox
8        image: docker.io/payalkharat/kubernetes-toolbox:latest
9        imagePullPolicy: IfNotPresent
10       env:
11         - name: DB_USERNAME
12           valueFrom:
13             secretKeyRef:
14               name: db-credentials
15               key: username
16       volumeMounts:
17         - name: secret-volume
18           mountPath: /etc/app/secrets
19           readOnly: true
20     volumes:
21       - name: secret-volume
22         secret:
23           secretName: db-credentials
```

4 . Verify:

```
kubectl exec secret-inspector -- cat /etc/app/secrets/username.
```

```
kubectl exec secret-inspector -- cat /etc/app/secrets/password
```

```
kubectl exec secret-inspector -- printenv DB_USERNAME
```

```
kubectl exec secret-inspector -- ls -l /etc/app/secrets
```

```

payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-3-SECRET-1$ kubectl delete pod secret-inspector
pod "secret-inspector" deleted from default namespace
payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-3-SECRET-1$
payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-3-SECRET-1$ ls
pod.yaml  secret.yaml
payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-3-SECRET-1$ kubectl apply -f secret.yaml
secret/db-credentials created
payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-3-SECRET-1$ kubectl get secret
NAME                TYPE      DATA      AGE
db-credentials       Opaque    2           4s
payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-3-SECRET-1$ kubectl apply -f pod.yaml
pod/secret-inspector created
payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-3-SECRET-1$ kubectl get pods
NAME                READY     STATUS      RESTARTS   AGE
secret-inspector    1/1       Running    0           5s
payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-3-SECRET-1$ kubectl exec secret-inspector -- cat /etc/app
/secrets/username
payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-3-SECRET-1$ kubectl exec secret-inspector -- cat /etc/app
/secrets/password
MyDBPassword123
payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-3-SECRET-1$
payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-3-SECRET-1$ kubectl exec secret-inspector -- printenv DB_
USERNAME
admin
payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-3-SECRET-1$ kubectl exec secret-inspector -- ls -l /etc/a
pp/secrets
total 0
lrwxrwxrwx    1 root    root          15 Aug 20 06:58 password -> ../data/password
lrwxrwxrwx    1 root    root          15 Aug 20 06:58 username -> ../data/username
payal_kharat@LAPTOP-V1NV0QFT:/mnt/d/CloudEthix_Internship/Kubernetes/POC/POC-3-SECRET-1$

```